Lecture 6: The Simplex Method

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Announcements

- Homework 1 due next Wednesday (2/15) at noon (12pm)

Recap

We studied some basic properties of linear optimization

- We observed that optimal solution tends to appear at extreme points
- To connect to the algebra, we defined the notion of basic solutions

To find a basic solution

1. Choose any m independent columns of A: $A_{B(1)}, ..., A_{B(m)}$
2. Let $x_i = 0$ for all $i \neq B(1), ..., B(m)$
3. Solve the equation $A\mathbf{x} = \mathbf{b}$ for the remaining $x_{B(1)}, ..., x_{B(m)}$.

- In a basic solution, there are no more than m positive entries
- There are finitely many basic solutions for a given LP

Recap: Basic Feasible Solutions

Definition

If a basic solution $\mathbf{x}$ also satisfies that $\mathbf{x} \geq 0$, then we call it a *basic feasible solution* (BFS).

Theorem

For the standard LP polyhedron $\{\mathbf{x} : A\mathbf{x} = \mathbf{b}, \mathbf{x} \geq 0\}$, *the followings are equivalent:*

1. $\mathbf{x}$ *is an extreme point*
2. $\mathbf{x}$ *is a basic feasible solution*

- This theorem connects the geometric property of LP to the algebraic property

Recap: Basic Feasible Solutions

Theorem

(LP fundamental theorem) Given a linear programming problem where A has full row rank m

- *If there is a feasible solution, there is a basic feasible solution;*
- *If there is an optimal solution, there is an optimal solution that is a basic feasible solution.*

Corollary

1. *In order to find an optimal solution, we only need to look among basic feasible solutions.*
2. *If an LP with m constraints (in the standard form) has an optimal solution, then there must be an optimal solution such that there is no more than m positive entries.*
3. *If an LP has a unique optimal solution, then it must be a BFS.*

Recap: Basic Feasible Solutions

We could enumerate all the BFSs to find the optimal solution. But it is not practical (too slow).

We want something smarter:

- Simplex method
- Inventor: George Dantzig

Idea of the Simplex method:

- Start from one BFS, either 1) find a neighboring BFS that can improve over the current one, or 2) stop.

We need to define more formally what it means by a neighboring solution

Neighboring BFS

Definition

Two basic solutions are *neighboring* (or adjacent) if they differ by exactly one basic (or non-basic) index

For example, a BFS constructed by using columns $\{1, 2, 3\}$ is a neighbor to the BFS constructed by using columns $\{1, 3, 5\}$, but not $\{1, 4, 5\}$.

Example

Consider the production planning problem:

$$
\begin{array}{rrrrrrrl}
\text{minimize} & -x_1 & -2x_2 & & & & & \\
\text{subject to} & x_1 & & +s_1 & & & & = 100 \\
& & 2x_2 & & +s_2 & & & = 200 \\
& x_1 & +x_2 & & & +s_3 & & = 150 \\
& x_1, & x_2, & s_1, & s_2, & s_3 & & \geq 0
\end{array}
$$

We can list the basic (feasible) solutions

Indices	$\{1,2,3\}$	$\{1,2,4\}$	$\{1,2,5\}$	$\{1,3,4\}$
Solution	(50, 100, 50, 0, 0)	(100, 50, 0, 100, 0)	(100, 100, 0, 0, -50)	(150, 0, -50, 200, 0)
Objective	-250	-200	infeasible	infeasible

Indices	$\{1,4,5\}$	$\{2,3,4\}$	$\{2,3,5\}$	$\{3,4,5\}$
Solution	(100, 0, 0, 200, 50)	(0, 150, 100, -100, 0)	(0, 100, 100, 0, 50)	(0, 0, 100, 200, 150)
Objective	-100	infeasible	-200	0

The other two choices $\{1,3,5\}$ and $\{2,4,5\}$ lead to dependent basic columns (therefore no basic solutions can be obtained)

Example

Indices	$\{1,2,3\}$	$\{1,2,4\}$	$\{1,2,5\}$	$\{1,3,4\}$
Solution	(50, 100, 50, 0, 0)	(100, 50, 0, 100, 0)	(100, 100, 0, 0, -50)	(150, 0, -50, 200, 0)
Objective	-250	-200	infeasible	infeasible
Indices	$\{1,4,5\}$	$\{2,3,4\}$	$\{2,3,5\}$	$\{3,4,5\}$
Solution	(100, 0, 0, 200, 50)	(0, 150, 100, -100, 0)	(0, 100, 100, 0, 50)	(0, 0, 100, 200, 150)
Objective	-100	infeasible	-200	0

Say we start from $\{3,4,5\}$ and only allow to move from one BFS to its neighbor (and also we move only if the objective value decreases). Then

- If we choose $\{1,4,5\}$ first, then we can move along

$$\{3,4,5\} \to \{1,4,5\} \to \{1,2,4\} \to \{1,2,3\}$$

- We can also start from $\{2,3,5\}$, which will give us

$$\{3,4,5\} \to \{2,3,5\} \to \{1,2,3\}$$

- Both get to the optimal solution in finite steps.

Checking Neighbors

The main question is how to find a neighboring BFS that improves the current one (reduces the objective function)

- We only need to show how to do it for one step. Then we can simply iterate the same methods (until we reach optimal).

A naive way is simply to check all neighbors of the current BFS:

- There are $m(n-m)$ potential neighbors for a BFS (choosing one index to leave, choosing one to enter). For each one, one needs to solve a linear equation of m variables
- This works, but is likely to be very slow

We want a more efficient way to do it

Simplex Method

First we assume that we have somehow found a BFS whose basis is $B(1), ..., B(m)$.

Define

$$A_B = [A_{B(1)}, A_{B(2)}, ..., A_{B(m)}]$$

and let A_N be the matrix consist of the non-basic columns of A.

By rearranging the order of variables, we can write $A = [A_B, A_N]$, $\mathbf{x} = [\mathbf{x}_B; \mathbf{x}_N]$, where $\mathbf{x}_B$ are the basic variables, and $\mathbf{x}_N$ are the non-basic variables.

By definition, we have

$$\mathbf{x}_B = A_B^{-1}\mathbf{b} \qquad \mathbf{x}_N = 0$$

Find a Neighboring BFS

Now we want to find a neighboring BFS.

- We want to select a non-basic variable x_j to *enter* the basis.
- This means, we want to increase x_j from the current BFS

To do this, we consider moving $\mathbf{x}$ (current BFS) to $\mathbf{x} + \theta\mathbf{d}$ where

1. $d_j = 1$
2. $d_{j'} = 0$ for all other non-basic indices

What constraints do we have on $\mathbf{d}$?

- We need to guarantee that the resulting solution $\mathbf{x} + \theta\mathbf{d}$ is still feasible, that is:

$$A(\mathbf{x} + \theta\mathbf{d}) = \mathbf{b} = A\mathbf{x}$$

i.e., $A\mathbf{d} = 0$.

Basic Directions

Now we write $\mathbf{d} = [\mathbf{d}_B; \mathbf{d}_N]$. Since we know that $d_j = 1$ and $d_{j'} = 0$ for all other non-basic indices. We have

$$A_B \mathbf{d}_B + A_j = 0$$

Therefore

$$\mathbf{d}_B = -A_B^{-1} A_j$$

That means, the entire $\mathbf{d}$ will be uniquely determined:

$$\mathbf{d} = [\mathbf{d}_B; \mathbf{d}_N] = [-A_B^{-1} A_j; 0; ...; 1; ...; 0]$$

where the 1 is at the jth entry. We call such $\mathbf{d}$ the jth basic direction.

Basic Directions

The jth basic direction only guarantees that the equality constraint $A\mathbf{x} = \mathbf{b}$ still holds. We also have the constraint that $\mathbf{x} \geq 0$. We need to verify that this still holds (i.e., $\mathbf{x} + \theta\mathbf{d} \geq 0$):

- For non-basic variables, it was 0 and the change in $\mathbf{d}$ is non-negative (either 1 or 0), therefore, it is still positive
- For basic variables, as long as they were strictly positive, there must exist a small θ such that $\mathbf{x} + \theta\mathbf{d} \geq 0$

Typically the basic variables of a BFS are all positive (i.e., m positive entries), there are cases that some basic variables in a BFS are equal to 0 (degeneracy), we will discuss those cases later

Change in the Objective Value

Remember the objective function of the original LP is $\mathbf{c}^T\mathbf{x}$. We can similarly decompose $\mathbf{c}$ into basic and non-basic parts, corresponding to the basic/non-basic indices, i.e.,

$$\mathbf{c} = [\mathbf{c}_B, \mathbf{c}_N]$$

Now we study what happens to the objective function when we move from the BFS $\mathbf{x}$ to $\mathbf{x} + \theta\mathbf{d}$. The change is $\theta\mathbf{c}^T\mathbf{d}$.

We have (for $\mathbf{d}$ to be the jth basic direction):

$$\mathbf{c}^T\mathbf{d} = c_j - \mathbf{c}_B^T A_B^{-1} A_j \doteq \bar{c}_j$$

We call $\bar{c}_j$ the *reduced cost* of variable x_j.

Reduced Costs

The reduced cost

$$\bar{c}_j = c_j - \mathbf{c}_B^T A_B^{-1} A_j$$

is a very important concept in the simplex method. It corresponds to the change of the objective value if one tries to change the basis.

- The first term is the cost per unit increase in the variable x_j
- The second term is the cost of the compensating change in the basic variables necessitated by the constraints $A\mathbf{x} = \mathbf{b}$

Reduced Costs

Given the current basis and the index that one wants to add to the basis (j), the reduced costs can be easily computed:

- A positive reduced cost means that incorporating j into the current basis will increase the objective function (not good);
- A negative reduced cost means that incorporating j will reduce the objective function (we want to go in that direction)

Thus the reduced costs are the indicators of where we want to go.

Question: What is the reduced cost of a basic variable? Or what is $\bar{c}_j$ when $j = B(i)$?

$$\bar{c}_{B(i)} = c_{B(i)} - \mathbf{c}_B^T A_B^{-1} A_{B(i)} = c_{B(i)} - c_B^T \mathbf{e}_i = c_{B(i)} - c_{B(i)} = 0$$

Therefore, the reduced costs for basic variables are zero ($\mathbf{e}_i$ is a vector with 1 at ith position and 0s otherwise).

Reduced Costs

Theorem (Stopping Criterion)

Consider a basic feasible solution $\mathbf{x}$ *associated with the basis* $B(1), ..., B(m)$ *and let* $\bar{\mathbf{c}}$ *be the corresponding vector of reduced costs. If* $\bar{\mathbf{c}} \geq 0$*, then* $\mathbf{x}$ *must be optimal.*

Remark

This theorem gives a stopping criterion to the simplex algorithm: We stop when all the reduced costs are non-negative.

- It also means that if one could not find a neighbor solution that is better, then one must have already achieved optimal solution.

Proof

Suppose at a certain basic feasible solution $\mathbf{x}$, all the reduced costs $\bar{\mathbf{c}} \geq 0$. Then consider any feasible solution $\mathbf{y}$, and $\mathbf{d} = \mathbf{y} - \mathbf{x}$. We know that since both $\mathbf{x}$ and $\mathbf{y}$ are feasible, we must have $A\mathbf{d} = 0$. We can further write it as:

$$A_B \mathbf{d}_B + \sum_{i \in N} A_i d_i = 0.$$

Here N is the set of non-basic indices. Thus

$$\mathbf{d}_B = -\sum_{i \in N} A_B^{-1} A_i d_i$$

Now we study the change of objective value, we have

$$\mathbf{c}^T \mathbf{d} = \mathbf{c}_B^T \mathbf{d}_B + \sum_{i \in N} c_i d_i = \sum_{i \in N} (c_i - c_B^T A_B^{-1} A_i) d_i = \sum_{i \in N} \bar{c}_i d_i$$

We know that all the $\bar{c}_i$'s are nonnegative, also all the d_i must be nonnegative too since $\mathbf{y} \geq 0$. Therefore, the objective value at $\mathbf{y}$ must be at least as large as the objective value at $\mathbf{x}$. $\square$

Example of Reduced Cost

Consider the production problem in standard form:

$$\begin{array}{rrrrrrrl} \text{minimize} & -x_1 & -2x_2 & & & & & \\ \text{subject to} & x_1 & & +s_1 & & & & = 100 \\ & & 2x_2 & & +s_2 & & & = 200 \\ & x_1 & +x_2 & & & +s_3 & & = 150 \\ & x_1, & x_2, & s_1, & s_2, & s_3 & & \geq 0 \end{array}$$

If we are at basis $\{1, 2, 3\}$ with BFS $(50, 100, 50, 0, 0)$. Then the reduced costs are

$$\bar{c}_4 = 0 - [-1, -2, 0] \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 1 & 0 \end{bmatrix}^{-1} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = 0.5$$

Similarly, $\bar{c}_5 = 1$. Therefore, the reduced costs are all nonnegative, thus the BFS is optimal.

Example of Reduced Cost Continued

$$
\begin{array}{rrrrrrrl}
\text{minimize} & -x_1 & -2x_2 & & & & & \\
\text{subject to} & x_1 & & +s_1 & & & & = 100 \\
& & 2x_2 & & +s_2 & & & = 200 \\
& x_1 & +x_2 & & & +s_3 & & = 150 \\
& x_1, & x_2, & s_1, & s_2, & s_3 & & \geq 0
\end{array}
$$

If we are at basis $\{1, 4, 5\}$ with BFS $(100, 0, 0, 200, 50)$. Then the reduced costs are:

$$
\bar{c}_2 = -2 - [-1, 0, 0] \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix} = -2
$$

Similarly, $\bar{c}_3 = 1$. Therefore including x_2 in the basis in the next step will reduce the objective value.

Change of Basis

Now we know that at certain $\mathbf{x}$, if the reduced costs $\bar{c}_j \geq 0$, then $\mathbf{x}$ is optimal. What if some $\bar{c}_j < 0$?

- Then it means that by bringing jth variable (non-basic) into the basis, we can decrease the objective value. Thus we want to go in that direction

Assume $\mathbf{d}$ is the jth basic direction with $\bar{c}_j < 0$. We know that going in this direction can reduce the objective. But how much can we go?

- We need to make sure that $\mathbf{x} + \theta\mathbf{d} \geq 0$.
- We also want to go as far as possible
- Therefore, we choose

$$\theta^* = \max\{\theta \geq 0 | \mathbf{x} + \theta\mathbf{d} \geq 0\}$$

Stepsize

Now we know that we can go as much as

$$\theta^* = \max\{\theta \geq 0 | \mathbf{x} + \theta \mathbf{d} \geq 0\}$$

without violating any constraints. There are two cases for θ^*

- If $\mathbf{d} \geq 0$, then $\theta^* = \infty$. In this case, one can go unlimitedly far without making the solution infeasible, while keeping the objective decreasing. Therefore, the original LP is unbounded
- If $d_i < 0$ for some i, then we can solve:

$$\theta^* = \min_{\{i | d_i < 0\}} \left(-\frac{x_i}{d_i} \right)$$

Since $d_i \geq 0$ for $i \in N$. We can also write it as:

$$\theta^* = \min_{\{i \in B | d_i < 0\}} \left(-\frac{x_i}{d_i} \right)$$

Moving to a New Basis

Assume θ^* is finite (otherwise the problem is unbounded), then we can move to another feasible solution

$$\mathbf{y} = \mathbf{x} + \theta^* \mathbf{d}$$

Assuming that $B(\ell) \in \{B(1), ..., B(m)\}$ is the index such that $\theta^* = -\frac{x_{B(\ell)}}{d_{B(\ell)}}$ (it is possible for multiple indices to achieve this, we will discuss this later). Then we must have

$$y_{B(\ell)} = x_{B(\ell)} + \theta^* d_{B(\ell)} = 0$$

Thus, the basic variable $x_{B(\ell)}$ has become zero, whereas the non-basic variable x_j becomes positive (equal to θ^*). Meaning the basis has changed to

$$B(1), ..., B(\ell - 1), j, B(\ell + 1), ..., B(m)$$

An Iteration of the Simplex Method

We start from a BFS $\mathbf{x}$ (with corresponding basis B)

1. We first compute the *reduced costs* $\bar{\mathbf{c}}$

$$\bar{c}_j = c_j - \mathbf{c}_B^T A_B^{-1} A_j$$

   - If no reduced costs is negative, then $\mathbf{x}$ is already optimal
   - Otherwise choose some j such that $\bar{c}_j < 0$

2. Compute the jth basic direction $\mathbf{d} = [-A_B^{-1}A_j; 0; ...; 1; ...; 0]$
   - If $\mathbf{d} \geq 0$, then the problem is unbounded, i.e., the optimal value is $-\infty$.
   - Otherwise, compute $\theta^* = \min_{i \in B, d_i < 0} \{-\frac{x_i}{d_i}\}$

3. Let $\mathbf{y} = \mathbf{x} + \theta^* \mathbf{d}$. Then $\mathbf{y}$ is the new BFS with index j replacing $B(\ell)$ in the basis, where $B(\ell)$ is the index attaining the minimum in θ^*. Objective value is changed by $\theta^* \mathbf{c}^T \mathbf{d} = \theta^* \bar{c}_j$.

4. Repeat these procedures

Simplex Iteration

In most of the cases, the objective value will strictly decrease after one simplex method iteration.

- If that is the case, then since the objective keeps decreasing and there is only a finite number of BFS, the simplex method must stop in a finite number of iterations.
- However, there is possibility of the objective function staying the same after one iteration. This may cause issue if not dealt properly.
- We will discuss such cases (degeneracy cases) in the next lecture.